

A Seginer theorem for random tensors with arbitrary iid integrable entries

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Abstract

For every fixed order $k \geq 3$, we prove that the expected injective norm of a tensor with iid mean-zero integrable entries is comparable to its largest expected Euclidean fiber norm, uniformly over all rectangular dimensions and entry laws. This proves the conjecture of Lucca and Pesenti recorded as TR-21. The proof extends Zhou–Zhu’s shared-class discrepancy method across magnitude levels and combines it with Lucca–Pesenti’s sparse-tail theorem and Seginer’s matrix comparison.

1 Main theorem and attribution

For a real tensor $T \in \mathbb{R}^{n_1 \times \cdots \times n_k}$, its injective norm is

$$\|T\|_{\text{inj}} = \sup_{\substack{\|x_j\|_2=1 \\ 1 \leq j \leq k}} \left| \sum_{i_1=1}^{n_1} \cdots \sum_{i_k=1}^{n_k} T_{i_1, \dots, i_k} \prod_{j=1}^k x_{j, i_j} \right|. \quad (1.1)$$

Let $\mathcal{F}_j$ denote the one-dimensional fibers in mode j : all indices except the j th are fixed. Define

$$F(T) = \max_{1 \leq j \leq k} \mathbb{E} \max_{f \in \mathcal{F}_j} \|T_f\|_2. \quad (1.2)$$

The maximum over modes in this definition is *outside* the expectation.

Theorem 1.1 (Seginer comparison for iid tensors). *Fix $k \geq 3$. There is a finite constant C_k , depending only on k , such that every tensor $T \in \mathbb{R}^{n_1 \times \cdots \times n_k}$, $n_j \geq 2$, with iid real mean-zero integrable entries satisfies*

$$F(T) \leq \mathbb{E} \|T\|_{\text{inj}} \leq C_k F(T). \quad (1.3)$$

The constant is independent of the dimensions and the common entry law; in particular, the law may depend on the dimensions. No finite-variance or higher-moment hypothesis is imposed.

Both expectations are finite: the injective norm and every fiber norm are bounded by the sum of the absolute values of the entries. The lower comparison holds pointwise with constant one. The upper comparison is proved in Sections 2–5. It answers the full integrable-law formulation of TR-21 [3] and proves Lucca–Pesenti’s Conjecture 1.8 [2].

Corollary 1.2 (Bernoulli entries at every density). *Let $X \in \mathbb{R}^{n_1 \times \cdots \times n_k}$ have iid Bernoulli(p) entries, where $k \geq 3$, $n_j \geq 2$, and $p \in [0, 1]$. For $W = X - \mathbb{E}X$,*

$$F(W) \leq \mathbb{E} \|W\|_{\text{inj}} \leq C_k F(W).$$

This is immediate from Theorem 1.1. A separate Bernoulli argument is retained in Appendix A; it also identifies a summation refinement of Zhou–Zhu’s deterministic heavy-tuple estimate.

1.1 Relation to prior work and the additional step

Seginer proved the corresponding comparison for iid mean-zero matrix entries [4, Corollary 2.2]. Lucca and Pesenti formulated its tensor extension and proved a comparison under moment-equivalence hypotheses [2, Assumption 1.6, Theorem 1.7, and Conjecture 1.8]. Their sparse-tail estimate, used below, is a separate result from that moment-restricted comparison.

The light-heavy decomposition goes back to the Kahn–Szemerédi method [1]. Zhou and Zhu developed this approach for sparse tensors [5] and subsequently introduced a simultaneous multibox discrepancy estimate that charges each shared coordinate class only once [6, Lemmas 4.4–4.5, Proposition 4.6, and Appendix A]. Their Lemma 5.1 supplies the joint-rate argument for dependent occupancy counts. Their stated Bernoulli concentration theorem assumes $np \geq c \log n$ [6, Theorem 2.1].

The additional step here is a *magnitude-labelled* discrepancy estimate (Lemma 2.1) and a simultaneous heavy-sum estimate (Proposition 3.1). Coordinate classes are charged once across all magnitude labels, while fiber and variance contributions are summed with squared amplitudes. These are extensions of the cited Zhou–Zhu arguments, not independent origins of the discrepancy method or its entropy bookkeeping. The proof then uses a deterministic scale grid, a quantile truncation, and symmetrization to obtain the full-law comparison. The matrix and rare-tail estimates are imported in the precise forms stated next.

1.2 External estimates used in the proof

Seginer’s matrix comparison. For an iid mean-zero integrable random matrix A , there is a numerical constant C_S such that

$$\mathbb{E}\|A\|_{\text{op}} \leq C_S \left(\mathbb{E} \max_i \|A_{i,:}\|_2 + \mathbb{E} \max_j \|A_{:,j}\|_2 \right). \quad (1.4)$$

This is [4, Corollary 2.2, with moment exponent one]. It is used only for the elongated rectangular regime in Section 5.1.

Lucca–Pesenti’s rare-tail estimate. Fix $k \geq 3$. For independent integrable entries U_i on $[n]^k$, $n \geq 2$, such that $\mathbb{P}(U_i \neq 0) \leq n^{-5/4}$ for every i ,

$$\mathbb{E}\|U - \mathbb{E}U\|_{\text{inj}} \leq A_k \mathbb{E} \max_i |U_i|, \quad (1.5)$$

where A_k depends only on k . This is [2, Theorem 1.4 and Corollary 4.3], with $\varepsilon = 1/4$. Neither identical distributions nor higher moments are required, so deterministic zero padding is permitted.

The two estimates above are not reproved. The discrepancy and heavy-sum adaptations are proved below; the elementary net and light-part arguments follow the framework of [6, Lemmas 4.1–4.2].

1.3 Notation and proof outline

Write $[n] = \{1, \dots, n\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. All logarithms are natural unless a base is shown, and $\log_+ t = \max\{0, \log t\}$. The letters C_k and $C_{k,R}$ may denote different constants at different occurrences, with only the indicated dependence. For $\mu > 0$ and $v \geq 0$, define the upper Poisson rate

$$I_\mu(v) = \begin{cases} v \log(v/\mu) - v + \mu, & v \geq \mu, \\ 0, & v < \mu. \end{cases} \quad (1.6)$$

For fixed v , this rate is nonincreasing in the reference mean μ .

Padding a rectangular tensor with zeros into $[n]^k$, where $n = \max_j n_j$, preserves its injective norm and all modewise fiber maxima. Sections 2–4 work with independent bounded entries that are identically distributed on the original rectangle and zero outside it. Section 5 combines that bounded-law estimate with (1.5), handles elongated formats using (1.4), and removes symmetry. Section 6 explains why the iid placement of magnitudes cannot be replaced by an arbitrary deterministic coefficient tensor.

2 Magnitude-labelled discrepancy

Fix $k \geq 3$ and $n \geq 2$. Consider independent entries Y_i on $[n]^k$, iid on an original rectangular subarray and zero elsewhere, with $|Y_i| \leq 1$. For $m \in \mathbb{N}_0$ put $v_m = 2^{-m}$ and $p_m = \mathbb{P}(v_m/2 < |Y_i| \leq v_m)$ on the original subarray. Labels with $p_m = 0$ can be discarded almost surely. In each mode choose pairwise disjoint, nonempty coordinate classes $D_j^\ell \subset [n]$, labelled by $\ell \in \mathbb{N}_0$. Write $Q_b = \prod_j D_j^{\ell_j}$ for $b = (\ell_1, \dots, \ell_k)$ and define

$$e_{b,m} = \#\{i \in Q_b : v_m/2 < |Y_i| \leq v_m\}, \quad \mu_{b,m} = p_m |Q_b|, \quad I_{b,m} = I_{\mu_{b,m}}(e_{b,m}).$$

The reference mean $\mu_{b,m}$ may exceed the actual mean after padding. For a finite nonempty collection $\mathcal{B}$ of pairs (b, m) , let

$$H(\mathcal{B}) = \sum_{j=1}^k \sum_{\ell \text{ used in mode } j} h(D_j^\ell), \quad h(S) = |S| \log \frac{en}{|S|},$$

$$c(b, m) = \log 2 + 2 \sum_{j=1}^k \log(\ell_j + 2) + 2 \log(m + 2).$$

Each used coordinate class contributes once to H , irrespective of the number of its coordinate-box or magnitude-label appearances. Set $H(\emptyset) = 0$; sums over an empty collection are zero.

Lemma 2.1 (Joint discrepancy with magnitude labels). *For $R > 0$ and $K = 2(R + k + 3)$, with probability at least $1 - n^{-R}$, simultaneously for every such assignment and finite collection,*

$$\sum_{(b,m) \in \mathcal{B}} I_{b,m} \leq K \left(H(\mathcal{B}) + \sum_{(b,m) \in \mathcal{B}} c(b, m) \right). \quad (2.1)$$

Proof. We adapt the shared-class union bound of [6, Lemmas 4.4–4.5] and the joint-rate argument of [6, Lemma 5.1] to box–magnitude pairs. Fix a valid assignment and collection. An entry belongs to at most one selected pair (b, m) : distinct coordinate boxes are disjoint and an entry has at most one magnitude level. For nonnegative $t_{b,m}$, independence across entries and $1 + u \leq e^u$ therefore give

$$\mathbb{E} \exp \left(\sum_{\mathcal{B}} t_{b,m} e_{b,m} \right) \leq \exp \left(\sum_{\mathcal{B}} \mu_{b,m} (e^{t_{b,m}} - 1) \right).$$

The counts at different magnitude levels need not be independent. The displayed estimate uses their exclusivity within each entry instead. For each subset $A \subseteq \mathcal{B}$ and $s_a \geq 0$, choose $u_a \geq \mu_a$ with $I_{\mu_a}(u_a) = s_a$ and apply the estimate with $t_a = \log(u_a/\mu_a)$, zero outside A . Markov's inequality yields

$$\mathbb{P}\{I_a > s_a \text{ for all } a \in A\} \leq e^{-\sum_{a \in A} s_a}.$$

Expanding the product using $e^{z/2} - 1 = \frac{1}{2} \int_0^\infty e^{s/2} \mathbf{1}_{\{z > s\}} ds$ and integrating this joint tail gives $\mathbb{E} \exp(\frac{1}{2} \sum_{\mathcal{B}} I_a) \leq 2^{|\mathcal{B}|}$. The probability that (2.1) fails for this assignment is thus at most

$$2^{|\mathcal{B}|} e^{-KH(\mathcal{B})/2} \prod_{a \in \mathcal{B}} e^{-Kc(a)/2}.$$

Put $Z_n = \sum_{\emptyset \neq S \subset [n]} e^{-Kh(S)/2}$. The bound $\binom{n}{s} \leq (en/s)^s$ and monotonicity of $s \log(en/s)$ on $[1, n]$ give

$$Z_n \leq \sum_{s=1}^n (en/s)^{-(K/2-1)s} \leq n(en)^{-(K/2-1)} \leq n^{-R-1} \leq 1.$$

For fixed $\mathcal{B}$, summing $e^{-KH/2}$ over assignments is at most $Z_n^q \leq Z_n$, where $q \geq 1$ is the number of used coordinate classes; dropping their disjointness restrictions only increases the sum. There is no coordinate-set choice for a magnitude label. The remaining union over finite nonempty collections is at most

$$Z_n \left[\prod_{b,m} (1 + 2e^{-Kc(b,m)/2}) - 1 \right] \leq Z_n (e^{S_K} - 1), \quad S_K = 2^{1-K/2} \left(\sum_{q \geq 0} (q+2)^{-K} \right)^{k+1}.$$

Since $K \geq 12$, $\sum_{q \geq 0} (q+2)^{-K} \leq 1$ and $S_K \leq 2^{1-K/2} \leq 1/32$, so $e^{S_K} - 1 < 1$. The countable union, or equivalently its increasing finite truncations, has probability at most $Z_n \leq n^{-R}$. $\square$

3 All magnitude levels in one heavy sum

The three-class decomposition below follows [6, Proposition 4.6]. Its modification is to sum all magnitude levels within each fiber, variance, and shared-class entropy estimate.

Proposition 3.1 (Simultaneous heavy-sum estimate). *Let Y be as in Section 2, and suppose that (2.1) holds with constant K for all class assignments and finite collections. Write*

$$\sigma^2 = \mathbb{E} Y_i^2 \quad \text{on the original rectangle,} \quad \Phi = \max_{j,f \in \mathcal{F}_j} \|Y_f\|_2.$$

For every $D \geq \max\{1, n\sigma^2, 4\Phi^2\}$ and every tuple of unit vectors $x_1, \dots, x_k$, put $q_i = \prod_j |x_{j,i_j}|$ and $\tau = \sqrt{D}/n$. Then

$$\sum_{i: |Y_i|q_i > \tau} |Y_i|q_i \leq \left(k4^{k-1} + e^2 4^{k+1} + \frac{28k}{3} K \right) \sqrt{D} + KB_k, \quad (3.1)$$

where $B_k = 2^k [2 \log 2 + 8(k+1)]$.

Proof. If $\tau \geq 1$ there are no heavy tuples. Otherwise fix unit vectors, write $q_i = \prod_j |x_{j,i_j}|$, and let $D_j^\ell = \{i : 2^{-\ell-1} < |x_{j,i_j}| \leq 2^{-\ell}\}$. Empty classes are omitted. With $a_j = 2^{-\ell_j}$, $s_j = |D_j^{\ell_j}|$, $\alpha_{j,\ell} = |D_j^\ell| 2^{-2\ell}$, and $a_b = \prod_j a_j$, one has

$$\sum_{\ell} \alpha_{j,\ell} \leq 4, \quad |Q_b| a_b^2 = \prod_j \alpha_{j,\ell_j}.$$

Every tuple with $|Y_i|q_i > \tau$ belongs to a pair with $\rho_{b,m} = na_b v_m / \sqrt{D} > 1$. Since $D \geq 1$, such pairs satisfy $m + \sum_j \ell_j < \log_2 n$ and form a finite collection. For these pairs put $\lambda_{b,m} = e_{b,m} / \mu_{b,m}$. The nonnegative heavy sum is at most $\sum a_b v_m e_{b,m}$. Let $\mathcal{B}$ denote these pairs, and split them into

$$\begin{aligned} \mathcal{C}_1 &= \{(b, m) \in \mathcal{B} : na_j^2 \leq \rho_{b,m} \text{ for some } j\}, \\ \mathcal{C}_2 &= \{(b, m) \in \mathcal{B} : \lambda_{b,m} \leq e^2 \rho_{b,m}\} \setminus \mathcal{C}_1, \\ \mathcal{C}_3 &= \mathcal{B} \setminus (\mathcal{C}_1 \cup \mathcal{C}_2). \end{aligned}$$

The squared fiber charge in $\mathcal{C}_1$. Fix j and the other coordinate labels, and write $P = \prod_{i \neq j} a_i$. The condition is $a_j \leq v_m P / \sqrt{D}$. The sets $D_j^{\ell_j}$ are disjoint. If $c_m(f)$ counts level- m entries in fiber f , then

$$\sum_m v_m^2 c_m(f) \leq 4 \sum_{i \in f} Y_i^2 \leq D.$$

It follows, summing the qualifying pairs, that

$$\begin{aligned} \sum_{\substack{\ell_j, m: (b, m) \in \mathcal{B} \\ a_j \leq v_m P / \sqrt{D}}} a_j P v_m e_{b, m} &\leq \frac{P^2}{\sqrt{D}} \sum_{f \text{ in the fixed other classes}} \sum_m v_m^2 c_m(f) \\ &\leq \sqrt{D} \prod_{i \neq j} \alpha_{i, \ell_i}. \end{aligned}$$

The sum on the right controls all the qualifying pairs because the classes $D_j^{\ell_j}$ are disjoint. Summing over the other labels and then j (overcounting a pair that qualifies for several modes is harmless) gives

$$\sum_{\mathcal{C}_1} a_b v_m e_{b, m} \leq k 4^{k-1} \sqrt{D}. \quad (3.2)$$

The combined variance charge in $\mathcal{C}_2$. Here

$$\frac{a_b v_m e_{b, m}}{\sqrt{D}} \leq e^2 \frac{n p_m v_m^2}{D} \prod_j \alpha_{j, \ell_j}.$$

Since $\sum_m p_m v_m^2 \leq 4\sigma^2$, summing all labels gives

$$\sum_{\mathcal{C}_2} a_b v_m e_{b, m} \leq e^2 4^{k+1} \sqrt{D}. \quad (3.3)$$

Shared entropy in $\mathcal{C}_3$. The defining inequalities are $1 < \rho_{b, m} < n a_j^2$ for every j and $\lambda_{b, m} > e^2 \rho_{b, m}$. Thus $I_{b, m} \geq e_{b, m} \log(e \rho_{b, m})$. For $u \geq 0$, let

$$\mathcal{C}_3(u) = \left\{ (b, m) \in \mathcal{C}_3 : u < \frac{a_b v_m}{\sqrt{D} \log(e \rho_{b, m})} \right\}.$$

Integrating (2.1) over these subcollections gives

$$\frac{1}{\sqrt{D}} \sum_{\mathcal{C}_3} a_b v_m e_{b, m} \leq K \int_0^\infty H(\mathcal{C}_3(u)) du + K \sum_{\mathcal{C}_3} \frac{c(b, m) a_b v_m}{\sqrt{D} \log(e \rho_{b, m})}. \quad (3.4)$$

The interval endpoint equals $\rho_{b, m} / [n \log(e \rho_{b, m})]$. Monotonicity of $t / \log(et)$ for $t \geq 1$ bounds the lifetime of each used class $D_j^{\ell_j}$, across *all* magnitude labels, by $a_\ell^2 / \log(en a_\ell^2)$, with $n a_\ell^2 > 1$. Since $h(D_j^{\ell_j}) = |D_j^{\ell_j}| \log(en / |D_j^{\ell_j}|)$, its integrated cost is at most

$$\alpha_{j, \ell} \left(1 + \frac{\log_+(1/\alpha_{j, \ell})}{\log(en 4^{-\ell})} \right).$$

For $t_\ell = n 4^{-\ell} > 1$, this is bounded by $2\alpha_{j, \ell} + 1/t_\ell$: use $\log_+(1/\alpha) \leq \log t + 1/(\alpha t)$. Since $\sum_{t_\ell > 1} 1/t_\ell < 4/3$, the entropy integral is at most $28k/3$. This is the entropy estimate from [6, Appendix A], with one lifetime for each shared class across all magnitude labels. It imposes no lower bound on $n\sigma^2$. For the label term, $\log(e\rho) > 1$ and $\log(q+2) \leq q+1$ give

$$\sum_{b, m \geq 0} c(b, m) 2^{-(\sum_j \ell_j + m)} \leq B_k := 2^k [2 \log 2 + 8(k+1)].$$

Combining (3.2)–(3.4) proves (3.1). In particular, no number of magnitude levels enters the bound. $\square$

4 Bounded symmetric entry laws

Theorem 4.1 (Bounded-law high-probability estimate). *Fix $k \geq 3$ and $R > 0$. Let the entries of Y be iid and symmetric on a rectangular subarray of $[n]^k$, $n \geq 2$, and zero elsewhere. Assume $|Y_i| \leq 1$, and set*

$$\sigma^2 = \mathbb{E}Y_i^2 \quad \text{on the original rectangle,} \quad \Phi = \max_{j,f \in \mathcal{F}_j} \|Y_f\|_2.$$

There is a constant $C_{k,R}$ such that

$$\mathbb{P}\left\{\|Y\|_{\text{inj}} > C_{k,R}(1 + \sqrt{n\sigma^2} + \Phi)\right\} \leq 2n^{-R}. \quad (4.1)$$

Proof. For fixed unit vectors $x_1, \dots, x_k$, put $q_i = \prod_j |x_{j,i_j}|$. For deterministic $D_0 \geq \max\{1, n\sigma^2\}$, the light sum has summands

$$Y_i \prod_j x_{j,i_j} \mathbf{1}_{\{|Y_i|q_i \leq \sqrt{D_0}/n\}}.$$

They are centered because they are odd functions of symmetric entries, independent, bounded by $\sqrt{D_0}/n$, and have total variance at most $\sigma^2 \leq D_0/n$. Bernstein's inequality gives

$$\mathbb{P}\{|L_Y| > B\sqrt{D_0}\} \leq 2 \exp\left(-\frac{nB^2}{2 + 2B/3}\right).$$

Choose a deterministic $1/(2k)$ -net $\mathcal{T}$ of the unit sphere with $|\mathcal{T}| \leq (1 + 4k)^n$, as in [6, Lemma 4.1]. Replacing one vector at a time gives

$$\|Y\|_{\text{inj}} \leq 2 \max_{(x_1, \dots, x_k) \in \mathcal{T}^k} |Y(x_1, \dots, x_k)|.$$

This inequality is applied to the full multilinear form, not to its truncated light part. Union over the product net and the grid

$$\mathcal{D}_n = \{2^q : 0 \leq q \leq \lceil \log_2(4n) \rceil\}.$$

This grid has $O(\log(en))$ elements and maximum in $[4n, 8n)$. With $B = B_{k,R}$ sufficiently large, all allowed light bounds hold with failure probability at most n^{-R} . After sampling, select the smallest grid value $D_0 \geq \max\{1, n\sigma^2, 4\Phi^2\}$. This exists because $\sigma^2 \leq 1$ and $\Phi^2 \leq n$, and it is less than twice that maximum. Combine Lemma 2.1, Proposition 3.1, and the net bound. If $D_0 \geq n^2$ the heavy set is empty and the same light bound suffices. The two exceptional events have total probability at most $2n^{-R}$. Since D_0 is less than twice $\max\{1, n\sigma^2, 4\Phi^2\}$ and $D_0 \geq 1$, the additive KB_k in (3.1) is absorbed into a constant times $\sqrt{D_0}$. This proves (4.1). $\square$

Remark 4.2. The grid has the fixed lower endpoint 1. Its corresponding additive amplitude term in (4.1) is retained, not discarded; Section 5 charges it to F after truncation and rescaling. The simultaneous deterministic grid is what permits a random final choice of D_0 without conditioning away centering or independence.

5 Proof of the full-law comparison

We first prove the upper bound for symmetric iid integrable entries, then reduce the centered case to it by symmetrization.

5.1 Elongated rectangular formats

Let Z have symmetric iid integrable entries in the original format. Write $n = \max_j n_j$, $V = \prod_j n_j$. If $n \geq \prod_{i \neq j} n_i$ for a longest mode j , flatten into an $n \times P$ iid matrix with $P \leq n$. Its maximal column norm is the mode- j fiber maximum. To compare rows and columns, couple iid squared entries along a line of length nP and partition it into blocks of length P and, separately, length n . Each shorter block intersects at most two longer ones. Hence the expected maximal row norm is at most $\sqrt{2}$ times the expected maximal column norm. Seginer's estimate (1.4) and $\|Z\|_{\text{inj}} \leq \|\text{flatten}(Z)\|_{\text{op}}$ give $\mathbb{E}\|Z\|_{\text{inj}} \leq C_S(1 + \sqrt{2})F(Z)$. The coupling compares the two marginal expectations and need not be the row and column partition of one matrix realization.

5.2 Quantile truncation and the rare component

It remains to consider $V > n^2$. Pad the tensor with zeros to $[n]^k$, without changing its norm or fiber maxima, and put

$$\theta = \inf\{t \geq 0 : \mathbb{P}(|Z_i| > t) \leq n^{-5/4}\}, \quad Z_i^{\text{rare}} = Z_i \mathbf{1}_{\{|Z_i| > \theta\}}, \quad Z_i^c = Z_i \mathbf{1}_{\{|Z_i| \leq \theta\}}.$$

The quantile is finite and, by right-continuity of the strict tail, $\mathbb{P}(|Z_i| > \theta) \leq n^{-5/4}$. Both components are symmetric, independent across entries, and integrable. Equation (1.5) gives

$$\mathbb{E}\|Z^{\text{rare}}\|_{\text{inj}} \leq A_k \mathbb{E} \max_i |Z_i^{\text{rare}}| \leq A_k F(Z).$$

If $\theta = 0$ this already handles Z . For $\theta > 0$, each $0 \leq t < \theta$ has strict-tail probability greater than $n^{-5/4}$. Independence gives

$$\mathbb{E} \max_i |Z_i| \geq t(1 - e^{-Vn^{-5/4}}).$$

Letting $t \uparrow \theta$ and using $V \geq n^2$, $n \geq 2$, yields

$$\theta \leq 2\mathbb{E} \max_i |Z_i| \leq 2F(Z). \tag{5.1}$$

5.3 The bounded component and its variance term

Let $\sigma_c^2 = \mathbb{E}(Z_i^c)^2$ and $\Phi_c = \max_f \|Z_f^c\|_2$. Apply Theorem 4.1 to $Y = Z^c/\theta$. On its failure event use $\|Z^c\|_{\text{inj}} \leq \theta n^{k/2}$. Taking $R = k + 2$ gives

$$\mathbb{E}\|Z^c\|_{\text{inj}} \leq C_k(\theta + \sqrt{n\sigma_c^2} + \mathbb{E}\Phi_c) + 2\theta n^{k/2-R}. \tag{5.2}$$

Monotonicity of each squared fiber energy implies $\mathbb{E}\Phi_c \leq \sum_j \mathbb{E} \max_{f \in \mathcal{F}_j} \|Z_f\|_2 \leq kF(Z)$. For the variance term put $S = \sum_{i \text{ original}} (Z_i^c)^2$ and $\mu = \mathbb{E}S = V\sigma_c^2$. Each summand lies in $[0, \theta^2]$, so independence gives $\mathbb{E}S^2 \leq \mu^2 + \theta^2\mu$. If $\mu \geq 8\theta^2$, the Paley–Zygmund inequality yields

$$\mathbb{P}(S \geq \mu/2) \geq \frac{\mu^2}{4\mathbb{E}S^2} \geq \frac{1}{4(1 + 1/8)} = \frac{2}{9}.$$

The longest mode has V/n fibers; its maximal squared fiber energy is at least nS/V . Therefore

$$F(Z) \geq \sqrt{n/V} \mathbb{E}\sqrt{S} \geq \frac{2}{9\sqrt{2}} \sqrt{n\sigma_c^2}.$$

If $\mu < 8\theta^2$, then $\sqrt{n\sigma_c^2} < \sqrt{8}\theta\sqrt{n/V} \leq \sqrt{8}\theta$, absorbed by (5.1). Every term in (5.2), including its failure contribution, is thus at most a constant depending only on k times $F(Z)$. The triangle inequality and the rare-component estimate therefore prove $\mathbb{E}\|Z\|_{\text{inj}} \leq C_k F(Z)$ for symmetric iid integrable laws.

5.4 Symmetrization and completion of the proof

For the centered iid tensor T in Theorem 1.1, let T' be an independent copy and set $Z = T - T'$. Its entries are symmetric iid and integrable. Conditional Jensen and $\mathbb{E}T' = 0$ give $\mathbb{E}\|T\|_{\text{inj}} \leq \mathbb{E}\|Z\|_{\text{inj}}$. For each mode, the triangle inequality applied before taking the maximum and expectation gives $F(Z) \leq 2F(T)$. The symmetric comparison therefore proves the asserted upper bound. The lower bound is pointwise: fix all but one argument to coordinate vectors and optimize over the remaining fiber.

Equivalently, the final reduction reads

$$\mathbb{E}\|T\|_{\text{inj}} \leq \mathbb{E}\|T - T'\|_{\text{inj}} \leq C_k F(T - T') \leq 2C_k F(T).$$

Renaming the constant proves Theorem 1.1.

6 Why iid magnitude placement matters

A layerwise triangle inequality would replace the fiber norm of the whole tensor by a sum of layerwise fiber scales. Such a sum need not be controlled without a loss depending on the number of layers. Zhou–Zhu’s deterministic-weight bound [6, Corollary 2.2], expressed through a maximum weight and a maximum sampling probability, does not supply the joint estimate needed here. The proof above instead sums squared amplitudes in (3.2) and (3.3), and charges each coordinate class once across all magnitude levels in (3.4).

The iid placement of magnitudes cannot simply be discarded. To see this, let $M = 2^{2d^2}$ and construct an order-three coefficient tensor A supported on M disjoint $d \times d$ all-one blocks, with disjoint first- and second-mode coordinate sets and a separate third-mode coordinate for each block. Pad with zeros into cubic format. Then

$$\max_f \|A_f\|_2 = \sqrt{d}.$$

For independent Rademacher signs ε_i , an individual block of $\varepsilon \odot A$ is all positive with probability 2^{-d^2} , and these events are independent across blocks. Thus

$$\mathbb{P}\{\text{some block is all positive}\} = 1 - (1 - 2^{-d^2})^M \longrightarrow 1.$$

On that event, choose normalized all-one vectors on the corresponding first- and second-mode blocks and the appropriate third-mode coordinate vector. Their evaluation is d , so

$$\mathbb{E}_\varepsilon \|\varepsilon \odot A\|_{\text{inj}} \geq d[1 - (1 - 2^{-d^2})^M].$$

Consequently, a uniform comparison with the largest fiber norm is false for arbitrary deterministic magnitudes, even after averaging signs. This does not contradict Theorem 1.1: its entries are iid, whereas this coefficient construction fixes a highly nonuniform support.

A A direct Bernoulli argument

This appendix proves Corollary 1.2 directly from the Zhou–Zhu discrepancy and heavy-tuple framework, without using (1.4) or (1.5). Its additional steps are the retained-denominator estimate below and a simultaneous choice of the degree parameter. The all-density comparison is a deduction here, not a theorem stated in [6].

A.1 The retained-denominator summation

For $t > 1$, define

$$S_k(t) = \sum_{\substack{\ell \in \mathbb{N}_0^k \\ t2^{-|\ell|_1} > 1}} \frac{2^{-|\ell|_1}}{1 + \log(t2^{-|\ell|_1})}.$$

Lemma A.1 (A dyadic summation bound). *For fixed k , $S_k(t) \leq C_k/(1 + \log t)$ for every $t > 1$.*

Proof. Write $u = \log_2 t$. Grouping by $s = |\ell|_1$ gives

$$S_k(t) = \sum_{0 \leq s < u} \binom{s+k-1}{k-1} \frac{2^{-s}}{1 + (u-s)\log 2}.$$

For $s \leq u/2$, the denominator is at least $1 + (\log t)/2$, and $\sum_{s \geq 0} \binom{s+k-1}{k-1} 2^{-s} = 2^k$. For $s > u/2$, use the denominator bound 1 and the geometric tail estimate

$$\sum_{s > u/2} \binom{s+k-1}{k-1} 2^{-s} \leq C_k(1+u)^{k-1} 2^{-u/2} \leq \frac{C'_k}{1+u}.$$

The last inequality follows from the boundedness of a fixed polynomial times an exponentially decreasing function, including bounded u . These two bounds prove the claim. $\square$

Set $N = \max_j n_j$, $L_* = 1 + \lfloor \log_2 N \rfloor$, and let $1 \leq D \leq N$ be a parameter. For a nonnegative tensor padded with zeros into $[N]^k$, assume every fiber has sum at most D . Suppose that the multibox discrepancy inequality of Zhou–Zhu [6, Lemma 4.5] holds with constant K , labels $0, \dots, L_* - 1$, and reference means $(D/N)|Q_b|$:

$$\sum_{b \in \mathcal{B}} I_{(D/N)|Q_b|}(e(Q_b)) \leq K(H(\mathcal{B}) + |\mathcal{B}| \log(eL_*)). \quad (\text{A.1})$$

Here $e(Q)$ is the sum in a box, I_μ is (1.6), and for coordinate classes D_j^ℓ and boxes $Q_b = \prod_j D_j^{\ell_j}$,

$$H(\mathcal{B}) = \sum_{j=1}^k \sum_{\ell \text{ used in mode } j} |D_j^\ell| \log \frac{eN}{|D_j^\ell|}.$$

For unit vectors, use the heavy threshold $\tau = \sqrt{D}/N$. The dyadic classes in [6, Proposition 4.6] have amplitudes $a_\ell = 2^{-\ell}$; classes that occur in a heavy tuple require at most L_* labels. In its difficult class $\mathcal{C}_3$, write $a_b = 2^{-|b|_1}$ and $\rho_b = Na_b/\sqrt{D} > 1$. The estimate preceding the simplification in [6, equations (4.15)–(4.17)] gives an *absolute*, unnormalized label-cost contribution of at most

$$K \log(eL_*) \sum_{b \in \mathcal{C}_3} \frac{a_b}{\log(e\rho_b)}.$$

Lemma A.1, with $t = N/\sqrt{D}$, bounds this by

$$\frac{C_k K \log(eL_*)}{1 + \log(N/\sqrt{D})} \leq C'_k K,$$

because $D \leq N$ implies $N/\sqrt{D} \geq \sqrt{N}$. The other terms in [6, proof of Proposition 4.6] are unchanged: equations (4.12) and (4.13) give respectively $k4^{k-1}\sqrt{D}$ and $e^2 4^k \sqrt{D}$, while (4.16) gives $(28/3)kK\sqrt{D}$. The entropy estimate in [6, Appendix A] does not require a lower bound proportional to $\log N$. Since $D \geq 1$, the resulting heavy sum is at most $C_{k,K}\sqrt{D}$.

Thus the proof yields the deterministic heavy bound for $1 \leq D \leq N$; the assumption $D \geq c \log N$ is unnecessary after this refinement. This does *not* mean that the final bound can use $D = Np$ below logarithmic degree: the fiber-degree hypothesis must still hold.

A.2 Probabilistic inputs with a random degree scale

It suffices to treat $0 < p \leq 1/2$; complementing Bernoulli entries handles $p > 1/2$, and $p = 0, 1$ are trivial. Pad X and $\mathbb{E}X$ with zeros into $[N]^k$, which preserves the injective norm and fiber maxima. Put

$$d = Np, \quad \Delta = \max_f \sum_{i \in f} X_i, \quad D = \lceil \max\{1, d, \Delta\} \rceil \in \{1, \dots, N\}.$$

Fix a failure exponent $R > 0$. The padded counts in disjoint boxes are independent binomials with means at most $p|Q|$. Their upper tails are bounded by the Poisson rate $I_{p|Q|}$: this follows either from their exponential moment bound or from monotonicity in the reference mean. Consequently, [6, proofs of Lemmas 4.4–4.5] give, with probability at least $1 - N^{-R}$,

$$\sum_{b \in \mathcal{B}} I_{p|Q_b|}(e(Q_b)) \leq K(H(\mathcal{B}) + |\mathcal{B}| \log(eL_*))$$

for every partial partition and nonempty box collection. The zero padding therefore does not require identical distributions in this input.

For fixed v , $I_\mu(v)$ is nonincreasing in $\mu > 0$: on $\mu < v$, its derivative is $1 - v/\mu \leq 0$, and it is zero when $\mu \geq v$. Since $D \geq Np$, the same event implies (A.1) for every allowed D . Every fiber has degree at most D by definition.

For the light part, take the deterministic product net of [6, Lemma 4.1]. For each deterministic integer $a \in [1, N]$ with $a \geq d$, use threshold $\tau_a = \sqrt{a}/N$. At a fixed net point, the centered summands are bounded by τ_a and have total variance at most $p \leq a/N$. Bernstein's inequality gives

$$\mathbb{P}\{|L_W| > B\sqrt{a}\} \leq 2 \exp \left\{ -\frac{NB^2}{2 + (2/3)B} \right\}.$$

Choose $B = B_{k,R}$ sufficiently large to union bound over at most $(1 + 4k)^{kN}$ product-net points and at most N integer parameters. The light estimate then holds simultaneously for all these parameters, outside an event of probability at most N^{-R} . Selecting $a = D$ is valid even though D is random.

The heavy part of the mean is bounded by

$$p \sum_{i: q_i > \sqrt{D}/N} q_i \leq \frac{pN}{\sqrt{D}} \sum_i q_i^2 = \frac{d}{\sqrt{D}} \leq \sqrt{D}, \quad q_i = \prod_{j=1}^k |x_{j,i_j}|.$$

The refined deterministic bound controls the nonnegative heavy part of X . Combining these estimates and the factor-two net approximation gives

$$\|W\|_{\text{inj}} \leq C_{k,R} \sqrt{D} \quad \text{outside an event of probability at most } 2N^{-R}. \quad (\text{A.2})$$

Since $|W_i| \leq 1$, the norm is always at most $N^{k/2}$. Hence

$$\mathbb{E}\|W\|_{\text{inj}} \leq C_{k,R} \mathbb{E}\sqrt{D} + 2N^{k/2-R}. \quad (\text{A.3})$$

A.3 Charging the degree scale to the actual fibers

Let $V = \prod_j n_j$ and $\lambda = Vp$. Write Δ_j for the maximum count in a mode- j fiber. A fiber containing m ones obeys

$$\|W_f\|_2^2 = m(1-p)^2 + (n_j - m)p^2 \geq m/4.$$

It follows that

$$\mathbb{E}\sqrt{\Delta} \leq \sum_{j=1}^k \mathbb{E}\sqrt{\Delta_j} \leq 2kF(W).$$

On the event of at least one nonzero entry of X , each mode has a centered fiber norm at least $1-p \geq 1/2$. Thus

$$F(W) \geq \frac{1}{2}(1 - (1-p)^V) \geq \frac{1}{2}(1 - e^{-\lambda}). \quad (\text{A.4})$$

If $d = Np \geq 1$, a fixed fiber in a longest mode has count $M \sim \text{Bin}(N, p)$. The Paley-Zygmund inequality gives

$$\mathbb{P}\{M \geq d/2\} \geq \frac{d^2}{4\mathbb{E}M^2} \geq \frac{d}{4(d+1)} \geq \frac{1}{8}.$$

Therefore $F(W) \geq \sqrt{d}/(16\sqrt{2})$.

If $\lambda \geq 1$, (A.4) bounds $F(W)$ below by a numerical constant, and $\sqrt{d} \leq CF(W)$ whether $d \geq 1$ or $d < 1$. Since

$$\sqrt{D} \leq \sqrt{2}(1 + \sqrt{d} + \sqrt{\Delta}),$$

we obtain $\mathbb{E}\sqrt{D} \leq C_k F(W)$. Choose $R = k + 2$ in (A.3); its residual is bounded by a constant and is absorbed by $F(W)$. This proves the upper bound in this regime.

If $\lambda < 1$, let M_{tot} be the total number of ones. The elementary estimate

$$\mathbb{E}\|W\|_{\text{inj}} \leq \mathbb{E}\|X\|_F + \|\mathbb{E}X\|_{\text{inj}} = \mathbb{E}\sqrt{M_{\text{tot}}} + p\sqrt{V} \leq \lambda + \lambda = 2\lambda$$

uses $\sqrt{m} \leq m$ for nonnegative integers. By (A.4) and $1 - e^{-\lambda} \geq (1 - e^{-1})\lambda$ on $[0, 1]$, $F(W) \geq c\lambda$. The upper bound follows here as well. The lower bound in Corollary 1.2 is immediate by fixing all but one argument to coordinate vectors. This completes the proof.

A.4 The critical density $p = 1/n$

In cubic format n^k , a fixed mode contains n^{k-1} independent $\text{Bin}(n, 1/n)$ fiber counts. If Δ_j is their maximum, then, for fixed k and $n \rightarrow \infty$,

$$\mathbb{E}\sqrt{\Delta_j} \asymp_k \sqrt{\frac{\log n}{\log \log n}}.$$

For completeness, Chernoff's bound

$$\mathbb{P}\{\text{Bin}(n, 1/n) \geq t\} \leq (e/t)^t$$

and a union bound, followed by summing the tail above a sufficiently large multiple of $\log n / \log \log n$, give the upper scale. For the lower scale, for integers $1 \leq t \leq n$,

$$\mathbb{P}\{\text{Bin}(n, 1/n) = t\} \geq \frac{1}{4}t^{-t}.$$

Choosing $t = \lfloor ((k-1)/2) \log n / \log \log n \rfloor$ makes the expected number of fibers with this count tend to infinity; independence gives occurrence with probability tending to one. A centered fiber with count m has squared norm $m(1 - 2/n) + 1/n$, which lies between $m/4$ and $m + 1/n$ for $n \geq 2$. Thus the expected centered fiber maximum has the same square-root scale. Consequently, as $n \rightarrow \infty$,

$$\mathbb{E}\|X - 1/n\|_{\text{inj}} \asymp_k F(X - 1/n) \asymp_k \sqrt{\frac{\log n}{\log \log n}}.$$

This benchmark also follows from [6, proof of Proposition 4.6] without the retained-denominator refinement: that proof gives a heavy bound $C_{k,K}(\sqrt{D} + \log(eL_*))$, and $\log(eL_*) = O(\log \log n)$ is smaller than $\mathbb{E}\sqrt{D} \asymp_k \sqrt{\log n / \log \log n}$ in this regime. This observation uses the random-degree and expectation reductions above; it is not a direct application of [6, Theorem 2.1].

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